SRI HARI K M

Robotics & Embedded Systems Engineer

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PROFILE

Final-year B.Sc Computer Science student specialising in embedded systems and robotics, with hands-on industry experience at a robotics firm in UGV development for defence applications, CAD modelling, PCB design, and 3D printing. Lead author of a research paper on AI-driven lunar water-ice detection. Combines a hardware-first engineering mindset with strong software fundamentals, and is motivated to build advanced robotic and aerospace systems.

EDUCATION

Bachelor of Science, Computer Science 2024 – 2027 (Expected)

Loyola College (Affiliated with the University of Madras), Chennai

CGPA: 7.38 / 10.0 (through 4th semester)

EXPERIENCE

Embedded Systems & Robotics Intern

KAL-M Robotics and Innovations Pvt Ltd *2025 – Present*

Defence Division – UGV Development (Indian Army Project)

Sole CAD and PCB designer for the company; manage in-house 3D printing facility supporting rapid prototyping across all divisions

- Engineered suspension system, mine casing, and chassis for a military-grade UGV prototype
- Innovated a new UGV tank chassis design integrating Ingenium motors, improving on the prior drivetrain layout and torque delivery
- Conducted R&D and field testing of UGV prototypes for performance and reliability validation
- Diagnosed and resolved a motor brake-disengagement issue on a CAN-bus-driven BLDC motor via UART configuration, after the team had been stuck for two weeks assuming brake control was only accessible through CAN

Industrial Division – Coal Dust Suppression System UGV (Client Project)

- Created engineering drawings for the Innocule mast system and coordinated fabrication and material procurement

Commercial Division – TheBoxingBot, STEM Kits, Self Balancing Bot

- Designed and fabricated PCBs in KiCad for two products - the STEM education kit and the self-balancing robot - both currently in operation
- Contributed to hardware development of a combat-style boxing robot sold as a STEM education kit
- Organised and ran competitive robotics arenas, delivering hands-on robotics experience to students

RESEARCH

Lunar Water Detection via Remote Sensing *(Lead Author, Group of 5)*

Led the research and writing on remote sensing methods for lunar water ice detection, including radar/SAR-based approaches.

SELECTED PROJECTS

RTL-SDR Radio Observatory *(Ongoing)*

Built a software-defined radio station with a custom Fusion 360-designed discone antenna; implemented Whisper-based ATC transcription, ACARS packet detection, and ADS-B decoding.

CubeSat – MK1 *(Competition Entry)*

Designed a 3D CubeSat model and subsystem layout for a satellite design competition, covering payload placement and structural feasibility.

Smart Solar Tracker with IoT Weather Station *(Completed)*

Designed a solar tracker with an integrated IoT weather station monitoring temperature, humidity, and light intensity to optimise panel positioning.

RC Aircraft - Scratch Build (Second Build Complete)

Scratch-built a Cessna 150 RC aircraft from EPS foam, achieving correct centre of gravity on the first attempt. First build was lost to a transmitter-mishandling incident during its maiden flight: a second build is complete and awaiting flight testing.

TECHNICAL SKILLS

Embedded Systems: ESP32, Arduino, Raspberry Pi

Hardware Design: PCB design (KiCad), 3D CAD & mechanical design (Fusion 360), 3D printing / rapid prototyping, sensor integration

Robotics & Systems: UAV/UGV systems, ROS fundamentals

Software: Python, basic AI/ML, IoT development

ACHIEVEMENTS & ACTIVITIES

- 1st Place, Paper Presentation – JUNO Edition III, Technotsav, Guru Nanak College (Feb 2026)
- 3rd Place, Sustainathon – IoT Splice National IoT Fest, Kristu Jayanti College (Feb 2026)
- Study Tour – Satish Dhawan Space Centre (SHAR), Sriharikota, organised by Explore The Space, an ISRO-registered body (Aug 2025)
- Co-organizer, Let's Talk Robotics - a robotics community; lead hands-on demo sessions explaining Robotics, RTL-SDR, Hack RF, UGV system, at IIT Research Park events